

Cloud Operating Systems

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Agenda

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- * Cloud Operating Systems
 - * Paradigm
 - * Architecture
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Introduction

- * Definitions:

- * Distributed System (DS)
- * Cloud



What is a
Cloud OS?



Why Do You Need A Cloud Operating System?

- * Proceed in use of computing services.
- * Clouds have the potential to provide novel compute capacity to the average user.
- * Also provides more facilities for IT professionals.



Why Do You Need A Cloud Operating System?

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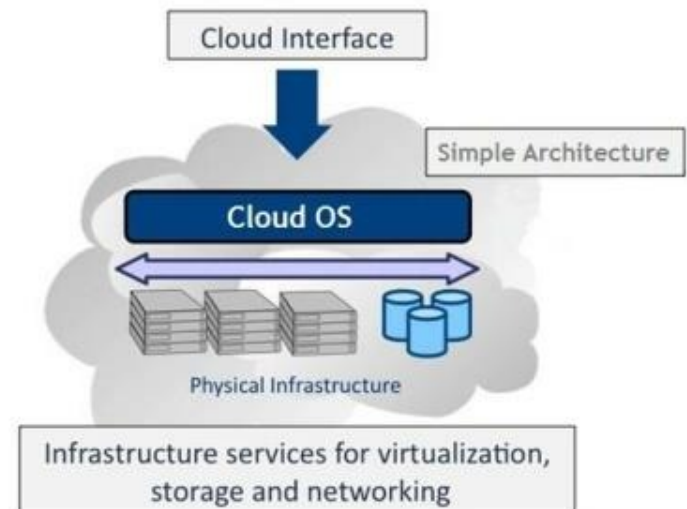
In order for the user to effectively control all of this computational power, We need an operating system for these new hardware platforms.



What is a Cloud Operating System?

* A simple definition:

The Cloud OS manages a number of servers and hardware devices, Giving the users the impression that they are interacting with a single infinite capacity and elastic cloud.

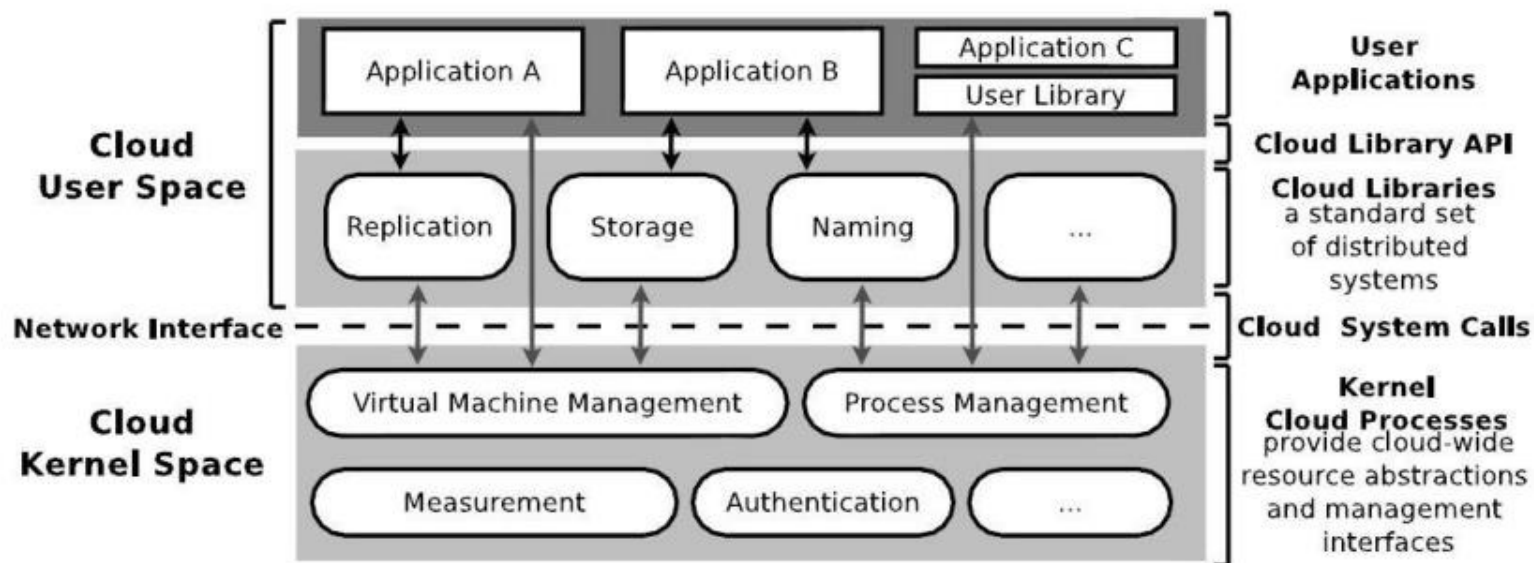


Paradigm

- * Paradigm used in Cloud Operating System:
 - * Object
 - * Thread
- * Object-Thread Model



Architecture



Common Challenges

- * Cloud computing infrastructure presents many common challenges with respect to the operating system. Such as:
 - * Scalability
 - * Elasticity of Demand
 - * Faults



Scalability

- * The number of computers being added by cloud computing providers has been growing at a vast rate, driven largely by user demand for hosted computing platforms.
- * Cloud OS's need to embrace scalability and make it a first order design constraint.



Elasticity of Demand

- * A major commonality between clouds is that the demand is not static.
- * Furthermore, the variability of demand is much higher than in previous systems and the amount of available resources can be varied over a much broader range.



Faults

- * Cloud applications usually share cloud resources with other users and applications in the cloud.
- * The likelihood of software faults in clouds are high, Due to the inherent difficulty of writing multithreaded applications.



Current Cloud Operating Systems

- * Available Cloud Operating Systems:
 - * VMware vSphere 4
 - * Ubuntu Enterprise OS
- * And Web-Based Cloud Operating Systems



Web-Based Cloud Operating Systems

- * iCloud
- * eyeOS
- * Glide OS
- * g.ho.st



Conclusion

- * Design goals and chief design features:
 - * *Support for object-thread computational model*
 - * *Network-transparent object invocation*
 - * *Persistent storage*
 - * *Sharing via objects*
 - * *Automatic load balancing*



References

- [1] F.Pianese, P.Bosch, A.Duminuco, N.Janssens, T.Stathopoulos, M.Steiner, “ Toward a Cloud Operating System,” Network Operations and Management Symposium Workshops (NOMS Wksp), 2010.
- [2] I. Foster, C. Kesselman, J. M. Nick, and S. Tuecke, “The physiology of the grid: An open grid services architecture for distributed systems integration,” in Open Grid Service Infrastructure WG, Global Grid Forum, 2002.
- [3] “Amazon EC2.” [Online] <http://aws.amazon.com/ec2>.
- [4] D. Nurmi, R. Wolski, C. Grzegorzczak, G. Obertelli, S. Soman, L. Youseff, and D. Zagorodnov, “The eucalyptus open-source cloud-computing system,” in Proc. of Cloud Computing and Its Applications, Oct. 2008.
- [5] F. Chang, J. Dean, S. Ghemawat, W. C. Hsieh, D. A. Wallach, M. Burrows, T. Chandra, A. Fikes, and R. E. Gruber, “Bigtable: A distributed storage system for structured data,” in 7th USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2006.
- [6] S. J. Mullender, G. van Rossum, A. S. Tanenbaum, R. van Renesse, and H. van Staveren, “Amoeba: A distributed operating system for the 1990s,” IEEE Computer, vol. 23, no. 5, pp. 44–53, 1990.
- [7] E. V. Hensbergen, N. P. Evans, and P. Stanley-Marbell, “A unified execution model for cloud computing,” in In Proc. of the 3rd ACM SIGOPS International Workshop on Large Scale Distributed Systems and Middleware (LADIS), 2009.



References

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- [8] E. M. Maximilien, A. Ranabahu, R. Engehausen, and L. C. Anderson, "Toward cloud-agnostic middlewares," in Proceedings of OOPSLA '09, (New York, NY, USA), pp. 619–626, ACM, 2009.
- [9] E. K. Lua, J. Crowcroft, M. Pias, R. Sharma, and S. Lim, "A survey and comparison of peer-to-peer overlay network schemes," IEEE Communications Survey and Tutorial, vol. 7, pp. 72–93, March 2004.
- [10] SETI@Home. [Online] <http://setiathome.berkeley.edu>.
- [11] G. Fox, D. Gannon, S.-H. Ko, S. Lee, S. Pallickara, M. Pierce, X. Qiu, X. Rao, A. Uyar, M. Wang, and W. Wu, Peer-to-Peer Grids. John Wiley and Sons Ltd, 2003.
- [12] I. Foster and A. Iamnitchi, "On death, taxes, and the convergence of peer-to-peer and grid computing," in In 2nd International Workshop on Peer-to-Peer Systems (IPTPS), pp. 118–128, 2003.
- [13] N. Drost, R. van Nieuwpoort, and H. Bal, "Simple locality-aware coallocation in peer-to-peer supercomputing," in Proceedings of the 6th IEEE/ACM CCGrid Symposium, 2006.
- [14] J. Ritter, "Why gnutella can't scale. No, really." [Online] <http://www.darkridge.com/jpr5/doc/gnutella.html>, Feb 2001.
- [15] J. Albrecht, R. Braud, D. Dao, N. Topilski, C. Tuttle, A. C. Snoeren, and A. Vahdat, "Remote Control: Distributed Application Configuration, Management, and Visualization with Plush," in Proceedings of the Twenty-first USENIX LISA Conference, November 2007.



Thanks..



Any Question?